

**National University of Singapore
School of Computing**

Semester 1, AY202627

CS4246/5446

Reinforcement Learning and Sequential Decision Making

Issue: 26 August 2026

Project Guidelines (Draft)

Important Dates

- **Project Proposal Submission:** 1800 Monday, 28 Sep 2026
- **Project Presentation:** Week of 10 Nov 2026
- **Project Report Submission:** 2359 Sunday, 15 Nov 2025

Project Team

You will work in teams of **5–6** members. Project team sign-ups will be available under the *People > Project Teams* tab on Canvas. *Your project team may include the same or different members from your Tutorial and/or Assignment Teams, and members can be from different tutorial sessions.* Each team member is responsible for the overall success of the team. Choose your teammates carefully to ensure effective collaboration.

Project Topic

The project is a core component of the course, giving you the opportunity to explore **reinforcement learning and sequential decision making** in AI with an emphasis on **responsible and impactful applications**. Topics should be *directly relevant to the course materials and integrate Responsible AI* considerations such as fairness, transparency, accountability, and trust.

Themes:

- **AI Assistant for X:**

Develop intelligent assistants to support decision making in specific domains, *such as* education, finance, health, well-being, elderly care, or customer service.
- **AI for Good:**

Apply AI to address broader societal challenges, such as sustainability, climate change, healthcare, social protection, or ESG-driven initiatives.
- **Other suggested directions:**
 - **Human–AI Collaboration** – systems where humans and AI jointly plan, learn, and act
 - **Agentic AI and Autonomous Decision Support** – AI agents coordinating tools, reasoning, and actions to achieve complex goals

- **Planning and Reasoning with Foundation Models** – integrating LLMs into decision-making pipelines
- **Multi-Agent Systems** – AI for negotiation, cooperation, and competition across interacting agents
- **AI for Scientific Discovery and Innovation** – accelerating breakthroughs in domains like drug discovery, climate modeling, or materials science
- **Responsible AI by Design** – aligning AI behavior with human values, ethics, and governance frameworks

Project Types

You may choose **one** of the following project types:

1. Exposition Project on Frontiers of AI Planning and Decision Making

2. Explore a key framework, algorithm, or emerging area in AI. Possible topics include agentic workflows, PPO and its variants, reinforcement learning with human feedback (RLHF) or AI feedback (RLAIF) and their variants, Chain/Tree of Thought, hierarchical or model-based RL, AI for scientific discovery, human–AI collaboration, neuro-inspired models (e.g., MAMBA, liquid networks), digital twins, or multi-agent systems. Briefly explain the method, key challenges, and significance. Include a *demonstration*—e.g., reproducing results or running experiments. Emphasize analysis and insights into theoretical or practical impact.

3. Application Project

Define and address an AI problem within a chosen theme. Apply or extend advanced techniques such as PPO, DQN, Actor–Critic, hierarchical planning or reinforcement learning, model-based RL, LLM-assisted reasoning and planning, imitation and inverse reinforcement learning, or strategic multi-agent decision making.

4. Competition Project

Solve an AI task in planning, games, or robotics. Formal entry is not required, but demonstrate an AI controller in a suitable environment (e.g., OpenAI Gym, MiniGrid, PettingZoo, PyBullet, etc.). See *Information Resources* below.

5. Theoretical / Methodological Research (for the Brave Souls)

Propose an original algorithm or method in planning or decision making. Possible topics include symbolic–RL integration, neuro-symbolic reasoning, causal inference with knowledge graphs, human-alignment-aware RL, scalable multi-agent coordination, foundation models as world models and planners, continual learning, or neuro-inspired planning. Focus on conceptual innovation and potential impact; implementation is optional but encouraged.

- *If interested, discuss UROP or PhD opportunities with me.*

Project Requirements

The project contributes **30%** of the final grade. Assessment is based on **the report and presentation**, but a *proposal must be submitted to receive a grade*.

a. Project Proposal¹

- **Length:** Maximum **1 page**, single-spaced, single-column, min. 10 point fonts, 1-inch margins (excluding title page & references).
- **Content:** Clearly outline your plan (not detailed specifics), providing background, problem statement, approach, team roles, deadlines, and risk assessment.
- **Submission:** Upload via **Canvas** with all team members' names, metric numbers, and email addresses.
- **Grading:** The proposal is **not graded**, but feedback will be provided.

b. Project Presentation

- **Duration:** ~15 minutes + 5 minutes Q&A (*subject to change*).
- **Format:** Present to the class or submit a recorded presentation if required.

c. Project Report

- **Writing Standard:** Academic, technical report.
- **Length:** Maximum **10 pages**, single-spaced, single-column, min. 12 point fonts, 1-inch margins (excluding title page & references). All important discussions, figures, tables, and materials must be included. Additional details (e.g., probabilistic graphics models, screenshots) can be placed in an appendix.
- **References:** Use a **Reference Manager** (e.g., Endnote, Zotero, Mendeley) and follow **Vancouver citation style**.
- **Submission:** Upload **PDF** via **Canvas** with all team members' details.

Audience

Your proposal, report, and presentation should be understandable to a **technical audience** familiar with AI but not necessarily your specific topic (e.g., classmates).

Collaboration

- Clearly define team roles in both the proposal and final report (*adjustments from the proposal are acceptable*).
- Ensure all team members understand the entire project.
- **DO NOT** copy-paste or merely rephrase printed/online materials.
- Cite all sources properly.
- If using ChatGPT or other LLMs, specify your contributions vs. AI-generated content. You won't be penalized for AI use, but transparency is required.

¹ Feel free to start preliminary discussion on your topic in the discussion forum before the proposal deadline

Grading Criteria²

Your project will be graded on:

- **Report:** Organization, difficulty, achievement, soundness, innovation, clarity.
- **Presentation:** Organization, clarity, and ability to answer questions.

Best Project Award

The best project will receive a **mystery prize** at the end of the course!

Information Resources

- A list of relevant information repositories is provided [HERE](#)³.
- For data sources, try to access public available datasets or APIs.

² Refer to the Appendix A for detailed evaluation criteria.

³ For the competitions, you don't have to actually enter the competitions, just make use of the programming environments

Appendix A: Project Evaluation Guidelines

- Problem Understanding and Formulation
 - Motivation and rationale
 - Problem definition
 - Innovativeness
- Knowledge and Technical Depth
 - Relevance and significance
 - Accuracy of knowledge applied
 - Level of achievements
- Methodology and Results
 - Methodology
 - *Literature survey and prior work (if applicable)*
 - *Implementation (if applicable)*
 - Analysis and insights
- Effort and Initiative
 - Evidence of effort
 - Evidence of independent and teamwork
 - Contributions of team members clearly stated
- Report
 - Relevance of content
 - Organization
 - 10-page length strictly enforced
 - Writing style
- Presentation
 - Relevance of content
 - Organization
 - Clarity
 - Time management
 - Q&A

Appendix B: AI Planning and Sequential Decision-Making Competitions and Benchmarks

These resources support hands-on work in automated planning, scheduling, sequential decision making, multi-agent coordination, reinforcement learning, and adaptive agents. They complement standard machine-learning projects by emphasising long-horizon decisions, coordination, uncertainty, and adaptation.

(Links and status verified 21 August 2026.) Competition schedules change, please consult the official pages. Completed competitions are included only where their environments, starter kits, benchmarks, or results remain useful for projects.

1. Recent and Recurring Competitions

Competition	Focus	Why It Is Relevant	Status and Link
International Planning Competition IPC 2026	Automated planning; the Flagship benchmark 2026 edition competition for included numeric planning (Optimal, Satisficing, and tracks) and first Epistemic Planning track	Agile planning and the reasoning about beliefs and knowledge in multi-agent settings	2026 edition completed; Numeric Planning tracks and Epistemic Planning track
Automated Negotiating Agents Competition ANAC 2026	Automated negotiation, bidding and acceptance strategies, opponent modelling, supply-chain negotiation, and human-agent negotiation	Directly applies decision theory and multi-agent strategy to autonomous agents	Recurring competition; 2026 edition completed; official site (ANAC Live Competition)
Flatland ECML 2026 Challenge	Multi-agent rescheduling, railway routing, coordination, intermediate stops, and time windows	Supports planning, search, optimisation, operations research, and RL in a dynamic transport setting	2026 edition completed; official site (Flatland Association)
RoboCup Rescue Simulation League	Disaster simulation and intelligent agents for search, rescue, coordination, and resource allocation	A long-running setting for multi-agent planning and decision making under uncertainty	Recurring RoboCup league; official site (RoboCup 2026 로보컵 2026)
League of Robot Runners 2026	Task assignment, multi-agent path finding, execution control, and stochastic delays	Integrates scheduling, lifelong path planning, coordination, and robust execution in warehouse-scale scenarios	2026 main round completed; official site (League of Robot Runners)
NS-Gym AAMAS 2026 Non-	Decision agents that detect, adapt to, and	Supports RL, online meta-learning, and	2026 competition framework completed;

Competition	Focus	Why It Is Relevant	Status and Link
Stationarity Challenge	recover from environmental changes	continual learning remains reusable; under non-stationarity	official site (NS-Gym)

2. Archived Competitions with Reusable Environments

Resource	Focus and Use	Status and Link
Multi-Agent Programming Contest / MASSim	Cooperative agents with limited local information, coordinated task completion, multi-agent programming, and norm-sensitive scenarios	Last competition held in 2022; no confirmed next date; official site (Multi-Agent Programming Contest)
MineRL / BASALT	Minecraft-based sequential decision making, sample-efficient RL, imitation learning, human feedback, and tasks without predefined reward functions	Last BASALT competition held at NeurIPS 2022; MineRL repository (Proceedings of Machine Learning Research)
Lux AI Challenge – Season 3	Multi-agent 1v1 strategy-game agents using programmed policies, imitation learning, RL, or meta-learning	Closed on 24 March 2025; environment and submissions remain useful for projects; Kaggle page (Kaggle)

3. Benchmark Environments — Not Competitions

Environment	Focus	Why It Is Relevant	Link
MiniGrid	Configurable grid worlds involving navigation, for planning, objects, keys, doors, curriculum learning, and mazes, and hierarchical grounded missions	Lightweight environment for RL, curriculum learning, and instruction following	Official documentation (GitHub)
PettingZoo	Cooperative, competitive, turn-based, and simultaneous-action environments for multi-agent environments	Maintained standard API reference for multi-agent RL and sequential decision making	Official documentation (PettingZoo)

Useful Competition Portals

For new competition cycles, use the following official discovery pages rather than generic conference homepages:

- [ICAPS Competition Archive](#) for automated planning and scheduling competitions. ([ICAPS](#))
- [AAMAS 2026 Competitions](#) for multi-agent planning, routing, adaptation, games, and coordination challenges. ([Cyprus Conferences](#))
- [NeurIPS 2026 Competitions](#) for the latest accepted machine-learning, agent, robotics, optimisation, and evaluation competitions. Sixteen competitions were announced on 28 July 2026; participation schedules vary by competition. ([blog.neurips.cc](#))